

A Practical Guide to TensorFlow

TOPIC -- TENSORFLOW

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Eager execution TensorFlow includes an "eager execution" mode, which means that operations are evaluated immediately as opposed to being added to a computational graph which is executed later. Code executed eagerly can be examined step-by step-through a debugger, since data is augmented at each line of code rather than later in a computational graph. This execution paradigm is considered to be easier to debug because of its step by step transparency.

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TensorFlow TensorFlow is Google Brain's second-generation system. Version 1.0.0 was released on February 11, 2017. While the reference implementation runs on single devices, TensorFlow can run on multiple CPUs and GPUs (with optional CUDA and SYCL extensions for general-purpose computing on graphics processing units). TensorFlow is available on 64-bit Linux, macOS, Windows, and mobile computing platforms including Android and iOS. Its flexible architecture allows for easy deployment of computation across a variety of platforms (CPUs, GPUs, TPUs), and from desktops to clusters of servers to mobile and edge devices. TensorFlow computations are expressed as stateful dataflow graphs. The name TensorFlow derives from the operations that such neural networks perform on multidimensional data arrays, which are referred to as tensors. During the Google I/O Conference in June 2016, Jeff Dean stated that 1,500 repositories on GitHub mentioned TensorFlow, of which only 5 were from Google. In March 2018, Google announced TensorFlow.js version 1.0 for machine learning in JavaScript. In Jan 2019, Google announced TensorFlow 2.0. It became officially available in September 2019. In May 2019, Google announced TensorFlow Graphics for deep learning in computer graphics.

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TensorFlow TensorFlow serves as a core platform and library for machine learning. TensorFlow's APIs use Keras to allow users to make their own machine-learning models. In addition to building and training their model, TensorFlow can also help load the data to train the model, and deploy it using TensorFlow Serving. TensorFlow provides a stable Python Application Program Interface (API), as well as APIs without backwards compatibility guarantee for JavaScript, C++, and Java. Third-party language binding packages are also available for C#, Haskell, Julia, MATLAB, Object Pascal, R, Scala, Rust, OCaml,

and Crystal. Bindings that are now archived and unsupported include Go and Swift.

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AutoDifferentiation AutoDifferentiation is the process of automatically calculating the gradient vector of a model with respect to each of its parameters. With this feature, TensorFlow can automatically compute the gradients for the parameters in a model, which is useful to algorithms such as backpropagation which require gradients to optimize performance. To do so, the framework must keep track of the order of operations done to the input Tensors in a model, and then compute the gradients with respect to the appropriate parameters.

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Distribute In both eager and graph executions, TensorFlow provides an API for distributing computation across multiple devices with various distribution strategies. This distributed computing can often speed up the execution of training and evaluating of TensorFlow models and is a common practice in the field of AI.

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Metrics In order to assess the performance of machine learning models, TensorFlow gives API access to commonly used metrics. Examples include various accuracy metrics (binary, categorical, sparse categorical) along with other metrics such as Precision, Recall, and Intersection-over-Union (IoU).

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TensorFlow is a software library for machine learning and artificial intelligence. It can be used across a range of tasks, but is used mainly for training and inference of neural networks. It is one of the most popular deep learning frameworks, alongside others such as PyTorch. It is free and open-source software released under the Apache License 2.0. It was developed by the Google Brain team for Google's internal use in research and production. The initial version was released under the Apache License 2.0 in 2015. Google released an updated version, TensorFlow 2.0, in September 2019. TensorFlow can be used in a wide variety of programming languages, including Python, JavaScript, C++, and Java, facilitating its use in a range of applications in many sectors.

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Social media Twitter implemented TensorFlow to rank tweets by importance for a given user, and changed their platform to show tweets in order of this ranking. Previously, tweets were simply shown in reverse chronological order. The photo sharing app VSCO used TensorFlow to help suggest custom filters for photos.

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Extensions TensorFlow also offers a variety of libraries and extensions to advance and extend the models and methods used. For example, TensorFlow Recommenders and TensorFlow Graphics are libraries for their respective functional. Other add-ons, libraries, and frameworks include TensorFlow Model Optimization, TensorFlow Probability, TensorFlow Quantum, and TensorFlow Decision Forests.

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TensorFlow 2.0 As TensorFlow's market share among research papers was declining to the advantage of PyTorch, the TensorFlow Team announced a release of a new major version of the library in September 2019. TensorFlow 2.0 introduced many changes, the most significant being TensorFlow eager, which changed the automatic differentiation scheme from the static computational graph to the "Define-by-Run" scheme originally made popular by Chainer and later PyTorch. Other major changes included removal of old libraries, cross-compatibility between trained models on different versions of TensorFlow, and significant improvements to the performance on GPU.

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LiteRT LiteRT, formerly known as TensorFlow Lite, has APIs for mobile apps or embedded devices to generate and deploy TensorFlow models. These models are compressed and optimized in order to be more efficient and have a higher performance on smaller capacity devices. LiteRT uses FlatBuffers as the data serialization format for network models, eschewing the Protocol Buffers format used by standard TensorFlow models.

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Edge TPU In July 2018, the Edge TPU was announced. Edge TPU is Google's purpose-built ASIC chip designed to run TensorFlow Lite machine learning (ML) models on small client computing devices such as smartphones known as edge computing.

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Google JAX is a machine learning framework for transforming numerical functions. It is described as bringing together a modified version of autograd (automatic obtaining of the gradient function through differentiation of a function) and TensorFlow's XLA (Accelerated Linear Algebra). It is designed to follow the structure and workflow of NumPy as closely as possible and works with TensorFlow as well as other frameworks such as PyTorch. The primary functions of JAX are: